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Eliminating Routine Flaring by Innovative Ideas - A Successful Journey Towards Net-Zero

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Abstract

Flaring has been a longstanding practice in the oil and gas industry. The global community recognizes routine flaring as a significant contributor to greenhouse gas emissions, air pollution, and resource wastage. Abu Dhabi National Oil Company (ADNOC) has committed to achieving net zero greenhouse gas emissions by 2050, aligning itself with the United Arab Emirates' national sustainability vision.

This paper presents the various in-house modifications and improvements carried out in one of the ADNOC Onshore oil fields reducing the carbon footprint and thereby supporting the company and the nation to achieve the net- zero target.

The following modifications were proposed and successfully implemented by the Asset Operations *Team*.

1. Nitrogen is introduced as a blanketing for the water separation tanks, disposal water tanks and various drain vessels replacing fuel gas.
2. Settle out startup logic was developed for gas injection and gas lift compressors avoiding depressurization of the compressors' holdup volume before the start up. (settle out condition: starting a compressor in the pressurized state after a trip / stop without a need to depressurize it).
3. Avoiding the flaring during the pigging operation of various transfer lines between Central Degassing Station (CDS) and Gathering Station including satellite field trunk line.

For all the individual modifications, numerous brainstorming sessions were carried out with all the stakeholders including technical experts. The Management of Change (MOC) process was adhered *to,* and original equipment vendors were involved. For each modification, a detailed plan was developed, and the resources were allocated accordingly. To minimize the cost, asset operations executed the proposals using the in-house resources.

These initiatives enhanced the plant's operational performance and reliability while reducing flaring by 101 million standard cubic feet (MMSCF) per year, which equates to a reduction of 6,120 tons of CO2 emissions annually.

Introduction of the Surface Facilities

Qusahwira (QW) Oil Field CDS receives well fluids from oil producer wells located in South Areas from Remote Degassing Station (GS-1) via the 20″ transfer line, from Remote Degassing Station (GS-2) via the 24″ transfer line, from Mender field via *the* 14″ transfer line of Mender CGS (Central Gathering Station). Qusahwira CDS also receives well fluids from the producer wells located near CDS area. The CDS is designed for a production capacity of 96 MBOPD. There are three Multi-Selector Manifolds (M-0201/0202/0203) in order to divert the flow of a well to Multi Phase Flow Meter (MPFM) for well testing.

The Production Manifold receives well fluid from all the three Multi-Selector Manifolds located in CDS, From the 20″ transfer line of Qusahwira GS-1 and 24″ transfer line of Qusahwira GS-2. The Well fluid transfer line from GS-1 is provided with Pig Launcher (LP-1301) at GS-1 and Pig Receiver (RP-1301) at Qusahwira CDS. The Well fluid transfer line from GS-2 is provided with Pig Launcher (RP-3901) at GS-2 and Pig Receiver (RP-1308) at Qusahwira CDS.

The Mender wells fluid is also connected to the production manifold through the 14″ transfer pipeline. Well fluid transfer line from Mender is provided with Pig Launcher (LP-1303) at Mender and Pig Receiver (RP-1303) at Qusahwira CDS. Two MPFM (FT-0201/0202) are also provided downstream of MSM in order to facilitate individual well testing.

From the inlet facilities, production fluids flow to Separation Train 1, 2 & 3. Each train has its dedicated Slug Catcher, Production Separator and MOL Booster pumps. Train 1 consists of only the Production separator V-0301 and MOL Booster Pumps. Two new Slug Catchers 58-01-SC-0302 and SC-0303 are installed for Train 2 and Train 3 to prevent slugs coming into the separators and to stabilize the flow into the separators. The gas and liquid streams of Slug Catchers are remixed together at the outlet and sent to the Production Separators.

In the Three-phase Production Separators (V-0301/0302/0303) gas, oil and water are separated. Separated gas from Production Separators (V-0301/0302/V-0303) flows on back pressure control to a common header to 3 reciprocating compressor trains for injection gas and to 1 reciprocating compressor train for lift gas. Let down station is provided to feed Gas lift load during the outage of Gas lift compressor from the gas injection header.

The Produced water from Production Separators (V-0301/0302/V-0303) are sent to the Produced Water Treatment and Disposal System for oil removal and sub-surface disposal by re-injection to Water Disposal Wells. (figure 1) The Produced Water Treatment and Disposal System is comprised of three water separation tanks (T-2401/2402/2403) and three water disposal tanks (T-2404/2405/2406). The Water Separation Tanks are equipped with internal oil skimmers to remove oil from the effluent water before it is pumped to the water disposal wells. The vapor from Produced Water Treatment and Disposal System routes the liberated gas to the Vapor recovery header for further processing and to avoid flaring.

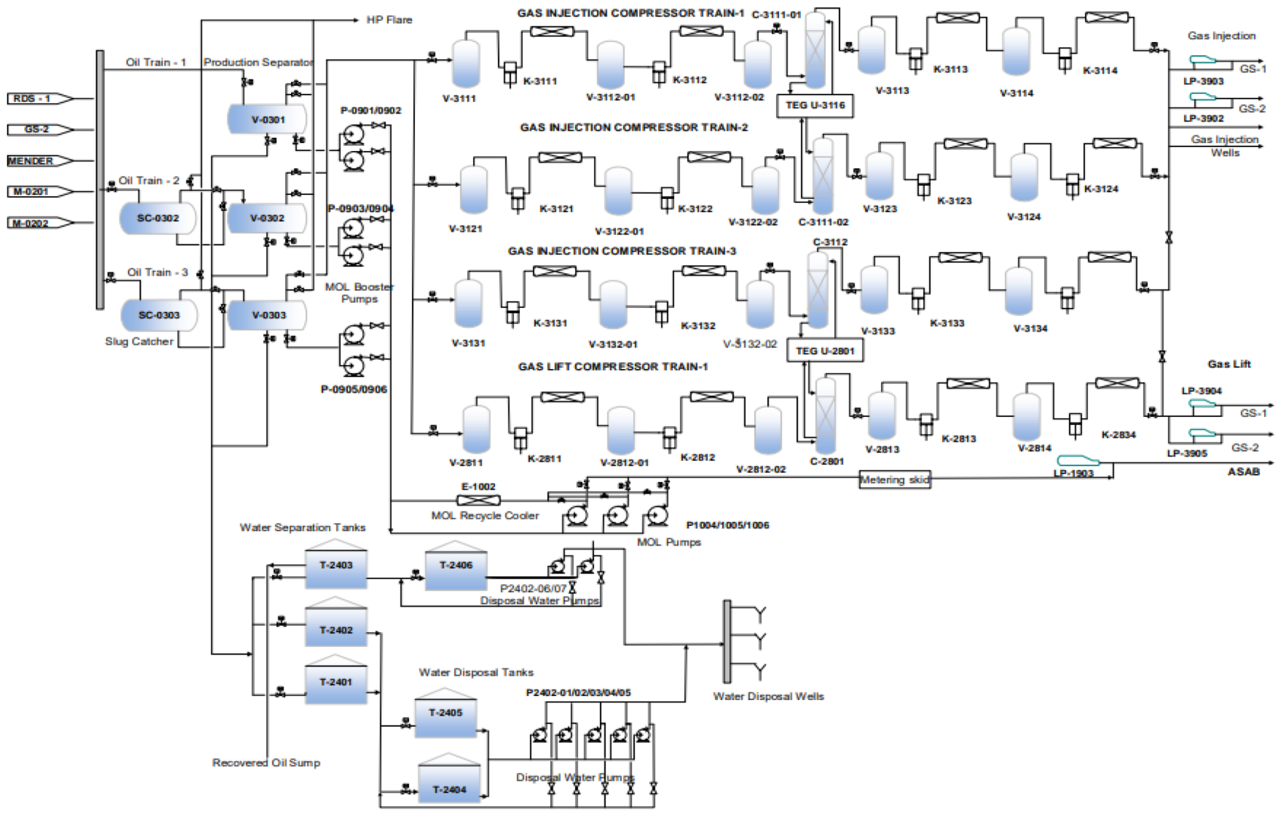

Figure 1—Schematic of QW CDS

The Crude Oil from Production Separators (V-0301/0302/0303) is routed towards the crude oil export system. The crude oil export pumps consist of two booster pumps (one operating with a spare) dedicated to each Oil Trains which feed to MOL pumps. The Oil from MOL Pumps is transferred to ASAB CDS through the 14″ pipeline for further treatment. The MOL transfer line is provided with Pig Launcher (LP-1303) at Qusahwira CDS and Pig Receiver (RP-1311) at ASAB CDS. (figure 2)

The Produced gas from Production Separators flows on back pressure control to a common header to 3 reciprocating compressor trains for injection gas and to 1 reciprocating compressor train for lift gas. Trains 1 and 2 are identical, having a capacity of 15 MMSCFD each. The three gas outlets from the production separators are gathered in a common header bringing the gas to each of the three compressor trains. All gas produced at the Qusahwira facility are either re-injected to maintain pressure on the reservoir or used in the production wells as lift gas to facilitate oil production. The final pressure at the injection gas wells is 240 bara (3500 psia) for gas injection and 145 bara (1500 psig) for lift gas at well head.

Each gas compressor train is provided with glycol contactor in between the LP and HP compressor stages to remove the moisture content and dry the gas to avoid hydrate formation in the downstream sections. Train-1 and Train-2, Compressors have a common glycol regeneration unit and similarly Train-3 and Gas lift compressors have its common glycol regeneration unit.

From CDS, the injection gas and Lift gas is distributed to GS-1 & GS-2 gas injection and Lift gas transfer lines, and to the CDS injection and lift gas wells. An Injection Gas Pig Launcher and Lift gas Pig Launcher is provided at CDS for the export transfer line to GS-1 and GS-2 to allow for periodic pigging operations. From GS-1 and GS-2, the injection gas and Lift gas is distributed to the gas injection wells and Lift gas wells connected to the respective pads well PADS.

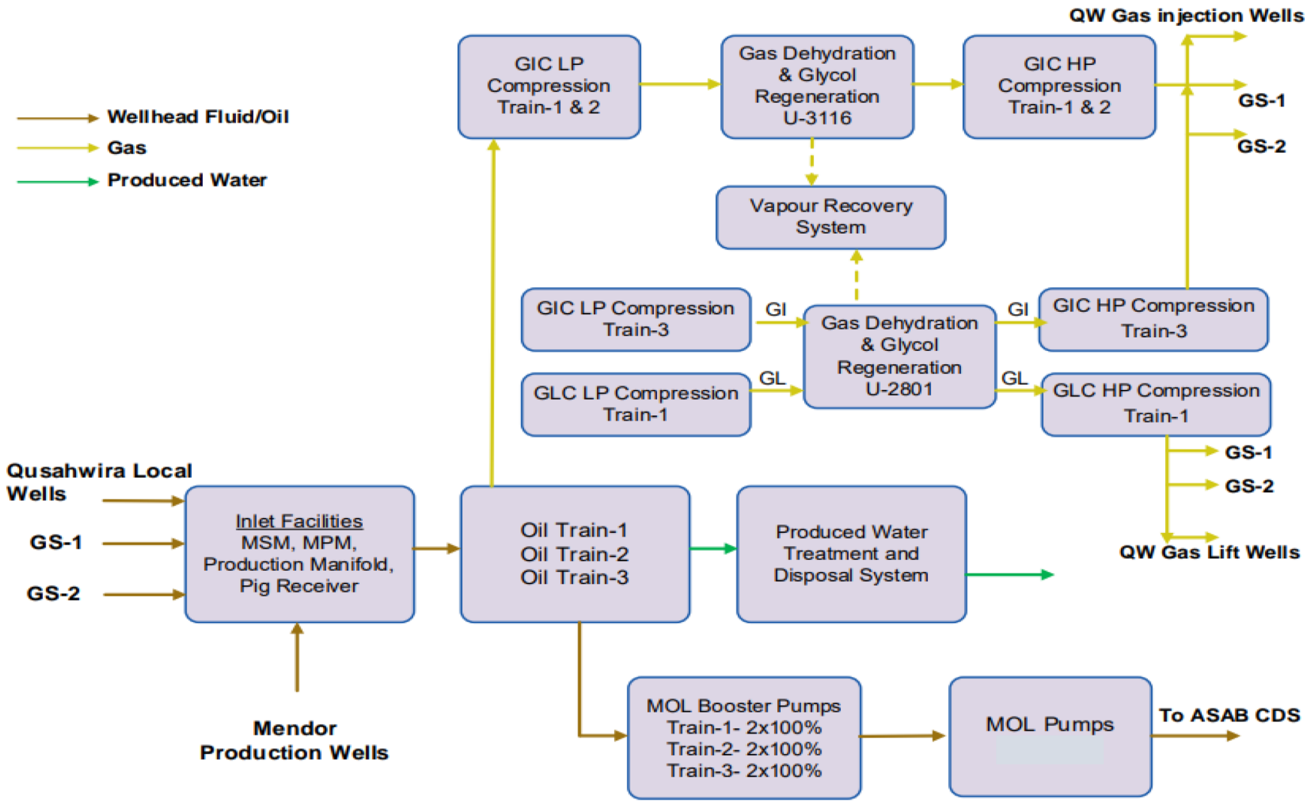

Figure 2—Block Diagram of QW CDS

Problem Statement

QW Field was having routine flaring of 0.27 mmscfd due to numerous challenges stated below.

Flaring from Tank Flare

QW CDS vapor recover header is fed from the water separation and Disposal tanks (6 Nos) outlet line from the tank blanketing system and from the drain vessels (5 Nos). Glycol regeneration system off gas header is also fed to the vapor recovery system. The vapor recovery header is routed to the tank flare system. The amount of gas flare in the tank flare was creating black smoke. Since QW field is in a strategic and environmentally sensitive area, it was very much required to control the flaring to reduce the greenhouse gas emission and its aftereffects like black smoke.

1. **Compressors Settle Out Start Up Logic:**
   Gas injection and Gas lift compressors were started normally from the initial start-up conditions as recommended by the vendors. Whenever the compressors tripped (a non-routine process upset scenario), all the stages were depressurized to their start-up condition which caused flaring the pressurized gas and it was delaying the compressor startup as the depressurization process took almost an hour. Since the volume of the gas being flared for every compressor trip is almost equal to 0.4 MMSCF in terms of depressurization of the compressor to the startup condition and 1.0 MMSCF in terms of compressor outage until bringing the compressor back to operational. It is evident from above flaring figures caused for any compressor trip; an innovative solution is required to minimize the flaring of greenhouse gases once again considering QW field's strategic high environmentally sensitive location.
2. **Pigging Operation:**
   Normal pigging operation is carried out for the transfer lines between QW GS-1 and GS-2 with the frequency of two times in a month for dewatering the line to avoid the corrosion and to maintain

the integrity of the transfer lines. The well fluid from the gathering station is transferred to central gathering station (CDS) using these 14″ and 24″ transfer line. The transfer lines have multi-phase flow in normal conditions. During the pigging operation, the back pressure in the line increases normally from 1.5 to 2 times of the normal operating pressure and enormous amount of gas will be accumulated just behind the pig. Once the pig is received at the receiver, The accumulated gas flows suddenly into the separator to maximum which cannot be handled by the available compressor capacity which caused flaring until the transfer line pressure is brought back to its normal condition which will be for one to two hours normally from the time of receiving the pig. It is not economically viable to design the compressor capacity to handle the temporary high volume of gas due to pigging operation. Since the pigging operation is a normal and unavoidable routing operation carried out in the oil and gas operations, in order to handle the huge volume of gas immediately after the pigging was a great challenge to avoid flaring of greenhouse gases which requires a robust, permanent solution without compromising the integrity of the transfer lines.

Literature

Flaring, the controlled burning of gaseous byproducts during oil extraction and processing, has been a longstanding practice in the oil and gas industry. While originally considered a necessary safety measure and a way to manage excess or non-commercially viable gas, flaring has come under increasing scrutiny due to its environmental impact. The global community now recognizes routine flaring as a significant contributor to greenhouse gas emissions, air pollution, and resource wastage. With ambitious net-zero targets being adopted across the globe, eliminating flaring has become a central focus for oil companies seeking to align with environmental, social, and governance (ESG) expectations.

Environmental and Health Impacts[1]

- Flaring contributes significantly to global warming and climate change. Methane, a potent GHG, is often incompletely combusted, leading to higher emissions than previously estimated.
- Over 250 toxins are associated with flaring emissions, affecting human health, biodiversity, and ecosystems
- Inefficient and unlit flares can result in up to **five times more methane emissions** than standard estimates suggest.

Major oil and gas operators, driven by stakeholder pressure and a desire to demonstrate climate leadership, have set ambitious goals to reduce or eliminate routine flaring. These commitments are often enshrined in sustainability reports, net-zero roadmaps, and ESG disclosures.

In response to global calls for environmental responsibility, ADNOC has committed to achieving net zero greenhouse gas emissions by 2050, aligning itself with the United Arab Emirates' national sustainability vision. This ambitious target reflects ADNOC's transition from traditional operational practices toward innovative solutions that mitigate environmental impact without compromising productivity. The company's journey involves a comprehensive review of existing flaring activities, a drive to adopt state-of-the-art emission reduction technologies, and the empowerment of field teams to pioneer efficiencies in daily operations. By fostering a culture of collaboration and continuous improvement, ADNOC leverages both technical expertise and stakeholder engagement to progress steadily toward its net zero aspirations, setting a benchmark for the region's energy sector.

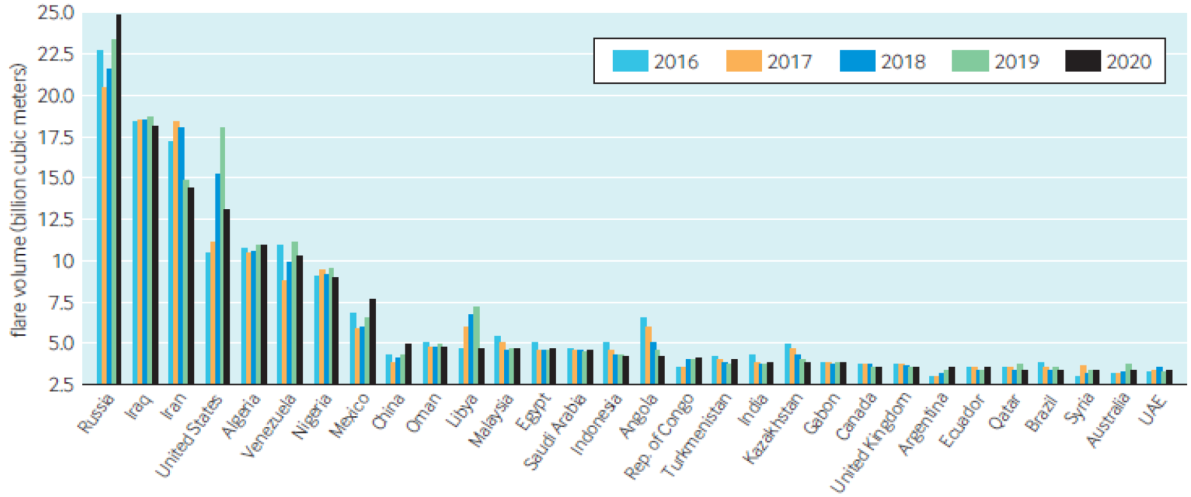

Figure 3—Flare volumes for the top 30 flaring countries, 2016-2020[1]

Economic and Strategic Considerations[2]

- Reducing flaring can save billions globally and improve energy security by utilizing wasted gas
- Strategic investments in infrastructure and technology are essential to transition toward low-carbon operations.

Completely eliminating flaring requires a multi-faceted approach, combining technology, operational excellence, and policy. Below are the principal strategies:

- Gas Capture and Utilization
- Operational Improvements
- Technological Innovations
- Policy and Market Mechanisms

Case Study: North Dakota, United States[3]

In the Bakken region, regulators required operators to capture a growing percentage of associated gas, ramping up from 74% in 2014 to over 91% today. This was achieved through aggressive investment in gathering infrastructure, gas processing plants, and enhanced coordination between oil and gas producers.

Case Study 3: Nigeria[4]

Nigeria, historically one of the world's largest flarers, has set ambitious targets for gas flare elimination. The Nigerian Gas Flare Commercialization Program (NGFCP) leverages public-private partnerships, regulatory reforms, and innovative technologies to transform waste gas into electricity and other valuable products.

Implementation Strategy

The cases mentioned in the problem statement were tackled as follows.

Nitrogen for Blanketing system

As per the original design of the plant, Fuel gas was used for the blanketing system for the tanks (as shown below in the DCS graphics, figure - 4) and drain vessels located inside CDS. For any reason, If the fuel gas pressure drops to 1.7 Bar, Nitrogen will be fed to the blanketing system with logic to open the SDV located in the nitrogen header and to close the Fuel gas SDV. This ensures the blanketing gas system never fails even during the total plant shutdown.

DCS Schematic of the fuel gas supply system is shown in the figure-4.

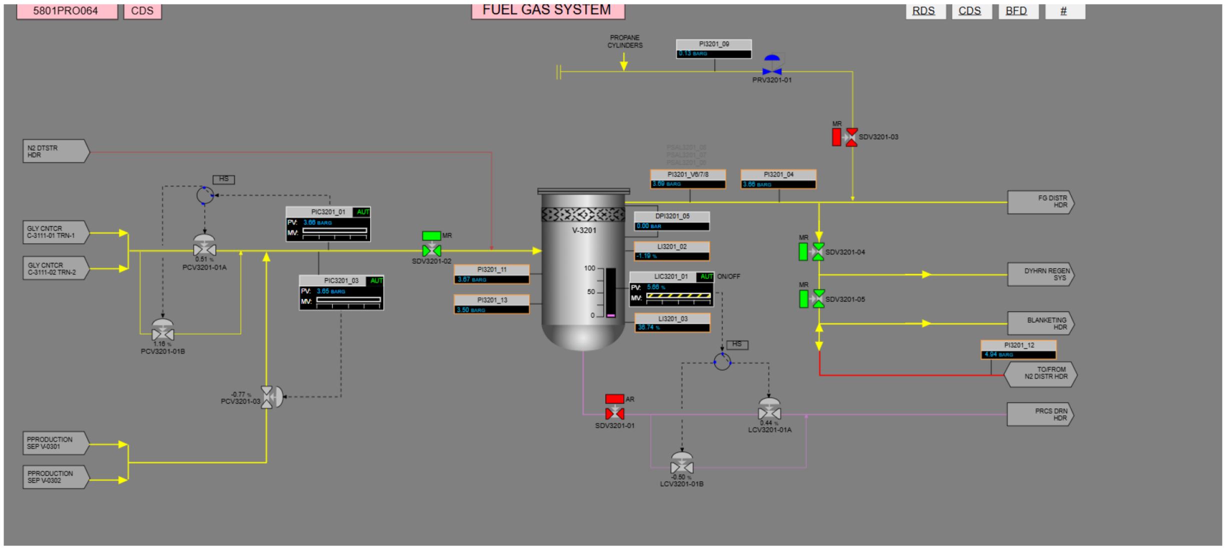

Figure 4—DCS Schematic of fuel gas distribution

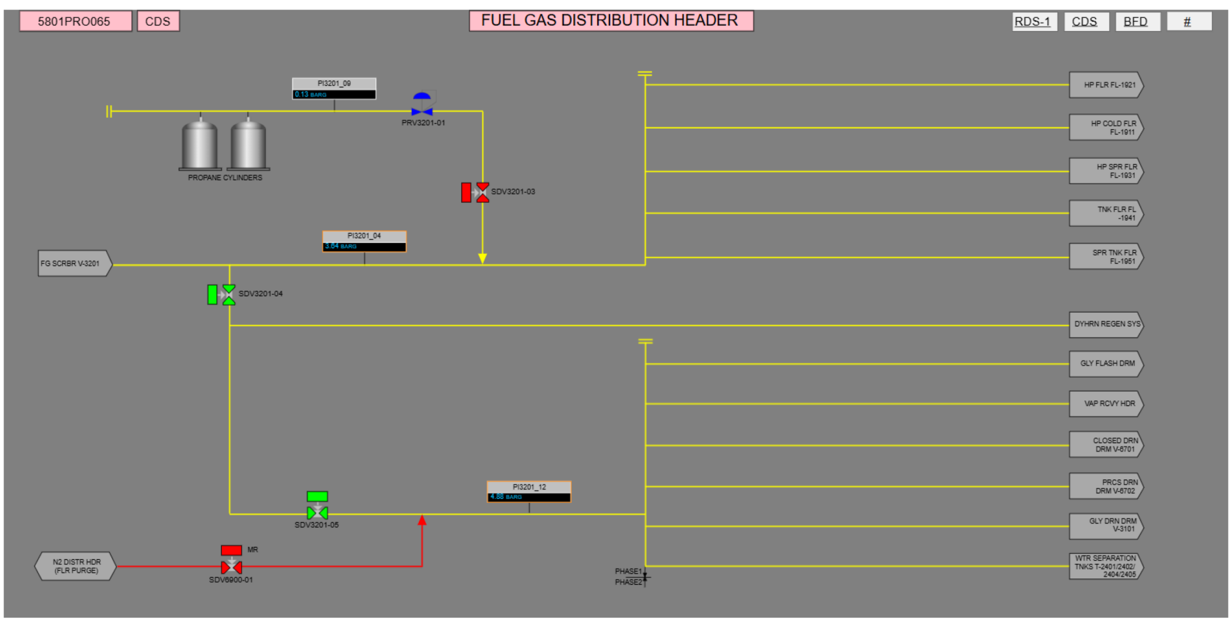

Figure 5—DCS Schematic of the Nitrogen back up of fuel gas

Since the fuel gas was used for the blanketing system, an amount of 71 MMSCF fuel gas was flared in the tank flare on a yearly basis. In order to reduce carbon emission, it was decided to use nitrogen as a primary source of blanketing system and to keep the fuel gas as a backup.

Before implementing, A detailed feasibility study was carried out for the impact on the Nitrogen system capacity and brainstorming sessions were carried out to optimize the nitrogen consumption to minimize the impact on the load on nitrogen skids.

Once the results were satisfactory, A detailed risk assessment session was carried out with all the stack holders and management of change (MOC) process was followed to get all the necessary approvals. Once the approval process is completed, the logic was changed in the DCS system by the inhouse team and there by the goal of reducing the emission by 60 MMSCF per year is achieved successfully.

Settle out startup of Gas Injection and Gas lift Compressors

QW CDS has three reciprocating compressors comprised of 4 stages to cater the gas injection requirement designed to compress 15 MMSCFD of dry gas and one four stage gas lift reciprocating compressor with a design capacity of 26 MMSCFD. Whenever any compressor tripped on ESD-3, the holdup volume was depressurized manually to the initial conditions before the startup. This procedure was time consuming which was causing not only to force to flare the holdup volume in the compressor while depressurizing but also to flare the produced gas due to non-availability of the compressor.

The possibilities of not depressurizing the compressor before startup was brainstormed with all the stakeholders. The merits and demerits were listed out. The feasibility was discussed with OEM vendors. After confirming that it is possible to start the compressor from its settle out pressure by keeping its integrity intact, a detailed automatic start up procedure was prepared to start the compressors from the settle out condition was prepared in house.

After getting approval from all the stakeholders, the startup logic was implemented in DCS by in-house resources and tested vigorously. (Figure - 6)

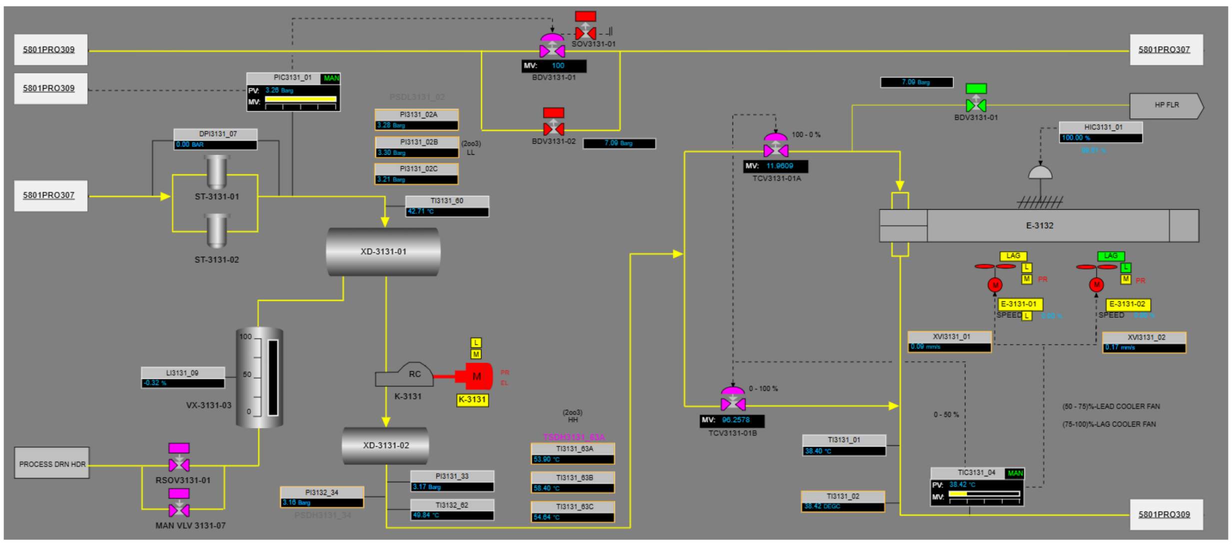

Figure 6—Compressor Auto Startup Sequence

Necessary training was given to all the operation and maintenance team and compressors started successfully from its settle out condition.

With the above logic modification, the flaring of the compressor inventory due to depressurization and reduction in the startup time considerably reduced the total flaring volume by 20 MMSCF and thus reducing the overall carbon footprint.

Avoiding Flaring During Transfer Line Pigging Operation

To reduce the flaring due to the pigging operation, a brainstorming operation was carried out among the operation section, and it was decided to use the Motor operated valve (MOV) provided in the production manifold to throttle, once the pig is received at the receiving end to release the gas slowly within the available compressor capacity without compromising the normal operating parameters.

The above point was discussed in detail with all the stack holders, and a risk assessment was carried out. After getting proper approval, the above-mentioned change in the pigging operation was implemented successfully and it was observed that the flaring due to the pigging operation was stopped completely.

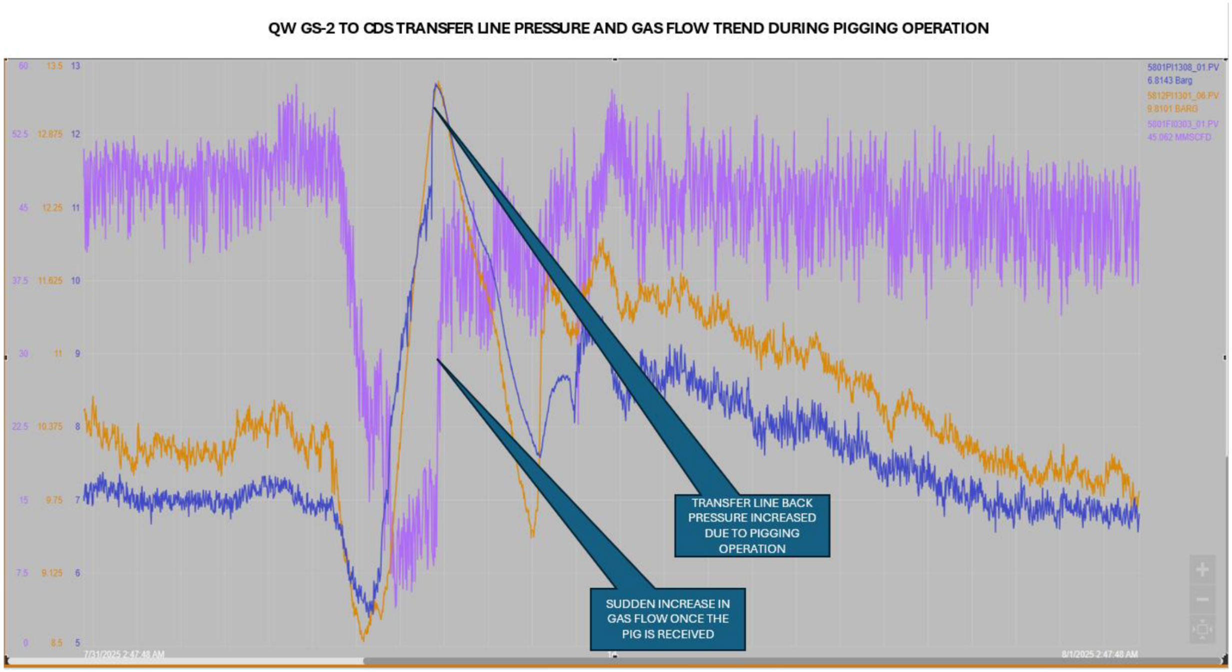

Figure 7—Pressure Trend During Pigging Operations

These changes in the operation, the flaring due to pigging amounting to 10 MMSCF, was eliminated, and the gas was conserved thereby reducing the carbon footprint.

Conclusion

For all the initiatives mentioned above, the actual reduction achieved is more than 100 MMSCF.

Idea	Flaring Reduction (mmscf)	CO2 Emission Reduction (Tons)
Nitrogen as Blanketing Gas	71	4302
Settle-out startup of Compressors	20	1212
Pigging Operation	10	606
Total	101	6120

The team exhibited their creativity and commitment to environmental and social responsibility by implementing innovative ideas. These initiatives enhanced the plant's operational performance and reliability while reducing flaring by 101 MMSCF per year, which equates to a reduction of 6,120 tons of CO_2 emissions annually.

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Glossary

CDS	Central Degassing Station
CO2	Carbon dioxide
DCS	Distributed Control System
ESD	Emergency Shutdown

GS	Gathering Station
HPSI	High Pressure Surface Injection
H_2S	Hydrogen Sulphide
HMI	Human Machine Interfaces
MMSCFD	Million Standard cubic feet per day.
MSDS	Materials Safety Data Sheets
MOV	Motor Operated Valve
P&ID	Piping and Instrument Diagram
PFD	Process Flow Diagram
PLC	Programmable Logic Controller
PRV	Pressure Relief Valve
PSV	Pressure Safety Valve
QW	QUSAHWIRA
RTU	Remote Telemetry Unit

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